Eg1 :- $3n+5 = O(n^2)$

Here $f(n) = 3n+5$ $g(n) = n^2$

$n^2 \rightarrow$ Highest order so $c = 3+5 = 8$.

based on $c = 8$, identify n_0

$c = 8$, $f(n) \leq c \cdot g(n)$

$3n+5 \leq 8 \cdot n^2$

$n_0 = 0$ $0+5 \leq 0$ ✗

$n_0 = 1$ $3+5 \leq 8$ ✓

$$\boxed{\text{So } n_0 = 1 \text{ \& } c = 8 \quad n \geq 1}$$

Eg2:- $3n+5 = O(n)$

Here $f(n) = 3n+5$ $\quad g(n) = n$

Here $f(n)$ & $g(n)$ order is same

so $c = 3+1 = 4$

$c=4$, $3n+5 \leq 4 \cdot n$

$n_0 = 0$, $5 \leq 0$ ✗

$n_0 = 1$, $8 \leq 4$ ✗

$n_0 = 2$, $11 \leq 8$ ✗

$n_0 = 4$, $17 \leq 16$ ✗

$n_0 = 5$, $20 \leq 20$ ✓

so $c = 4$

$n_0 = 5$

$n \geq 5$

Eg 3:-

$$3n - 5 = \Omega(n)$$

Here $f(n) = 3n - 5$

$g(n) = n$

$f(n) \geq c\,g(n)$

Here $f(n)$ & $g(n)$ same order.

so $c = 3 - 1 = 2$

When $c = 2$, $3n - 5 \geq 2n$.

$n_0 = 0$, $-5 \geq 0$ ✗

$n_0 = 1$ $-2 \geq 2$ ✗

$n_0 = 2$ $1 \geq 4$ ✗

$n_0 = 3$ $4 \geq 6$ ✗

$n_0 = 4$ $7 \geq 8$ ✗

$n_0 = 5$ $10 \geq 10$ ✓

So $c = 2$
$n_0 = 5$
$n \geq 5$